TÉCNICO LISBOA

Aprendizagem 2026/27

Homework II

Deadline 06/11/2026 (Friday) 23:59 via Fenix as PDF

- Submit Gxxx.ZIP in Fenix where xxx is your group number. The ZIP should contain two files: Gxxx_report.pdf with your report and Gxxx_notebook.ipynb.
- It is possible to submit several times on Fenix to prevent last-minute problems. Yet, only the last submission is kept.
- Exchange of ideas is encouraged. Yet, if copy is detected after automatic or manual clearance, homework is nullified and IST guidelines apply for content sharers and consumers, irrespectively of the underlying intent.
- Please consult the FAQ before posting questions to your faculty hosts.

I. Pen-and-paper [10v]

Consider the following dataset

D	y_1	y_2	y_3	y_4	y_5	y_6
x_1	0.54	0.74	1	0	0	N
x_2	0.49	0.98	0	1	0	N
x_3	0.49	0.56	1	1	1	P
x_4	0.42	0.46	1	0	0	N
x_5	0.68	0.33	1	1	0	P
x_6	0.50	0.46	1	1	0	P
x_7	0.37	0.84	1	1	0	P
x_8	0.40	0.66	1	0	0	N

and the problem of classifying observations as positive (P) or negative (N), according to the target variable y_6.

1) Consider $x_1 - x_6$ to be training observations, $x_7 - x_8$ to be testing observations, and $y_1 - y_5$ to be input variables.
    a. [2.0v] Learn a Bayesian classifier using the training observations assuming:

(i) $\{y_1, y_2\}$, $\{y_3\}$, and $\{y_4, y_5\}$ sets of independent variables (e.g., $y_1 \perp\!\!\!\perp y_3$ yet y_1 is not independent of y_2), and (ii) $y_1 \times y_2 \in R^2$ is normally distributed. Show all parameters (distributions and priors for subsequent testing).

b. [1.5v] Under a MAP assumption, classify each testing observation, showing all your calculus.

c. [2.5v] Explain the effects of the independence assumption (i) made in a. in terms of the parameters of the model and its generalization capabilities. Would there be any change in b. if a ML assumption was made, instead? Justify your answer and state one possible advantage of working under such an assumption.

2)

a. [2.5v] Let $y_3 - y_5$ be the input variables.

i. [1.5v] Compute the accuracy of a kNN with $k = 3$ and Hamming distance using a leave-one-out evaluation schema. Show all your calculus.

Note: For a given instance, if there is no radius containing only 3 neighbours, consider $k = 2$. For example, if an observation is at distance 0 from two observations, and at distance 1 from three others, taking 3 neighbours would require choosing among the three tied observations; instead, use only the two at distance 0.

ii. [0.5v] Would the accuracy change if the **Euclidean** distance was used over the same input variables? Justify your answer without repeating the computations.

iii. [0.5v] And if the **Chebyshev** distance, $d_\infty(x_i, x_l) = max_j|x_{ij} - x_{lj}|$, was used? Justify. What do your answers in ii. and iii. suggest about the choice of the order of a Minkowski distance when all input variables are Boolean?

b. [1.5v] Suppose again that $x_1 - x_6$ are training observations and $x_7 - x_8$ are testing observations, but that the target variable is now the numeric variable y_1. Adapt the kNN of a.i. to predict y_1 by replacing the majority vote with the mean of the y_1 values of the neighbours. Using the neighbourhoods of x_7 and x_8 already found in a.i. (no new distances need to be computed), predict y_1 for both testing observations and evaluate the predictions with the mean absolute error and the root mean squared error:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right|, \qquad RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}.$$

Note: This exercise is a small preview of regression tasks and a reminder that every classifier model studied in this course can be adapted to solve such a problem, including Naïve Bayes (as you will learn in lecture) and Decision Trees.

II. Programming and critical analysis [10v]

Use the heart.csv dataset available at the course's webpage and, using sklearn, assess the predictive models along this section. The dataset describes 284 patients who were examined for heart disease. The target `outcome` is 1 if the patient has heart disease (the positive class) and 0 otherwise. The 9 input variables are:

variable	meaning
`age`	age in years
`sex`	1 = male, 0 = female
`chest_pain`	type of chest pain: 1 = typical angina, 2 = atypical angina, 3 = non-anginal pain, 4 = asymptomatic (no chest pain). *Angina* is chest pain caused by insufficient blood flow to the heart muscle
`trestbps`	blood pressure at rest (mm Hg)
`fasting_blood_sugar`	1 if the blood sugar after fasting is above 120 mg/dl (a sign of diabetes), 0 otherwise
`restecg`	electrocardiogram (ECG) at rest: 0 = normal, 1 = ST-T wave abnormality, 2 = enlarged left ventricle
`thalach`	maximum heart rate reached during an exercise test (beats per minute)
`exercise_induce_angina`	1 if the exercise test caused angina, 0 otherwise
`oldpeak`	how much the ST segment of the ECG drops during exercise compared with rest; larger values suggest that the heart is not receiving enough blood

1) Using a 5-fold stratified cross-validation with shuffling (`random_state=0`), compare the performance of a kNN with $k = 5$ and a Naïve Bayes with Gaussian assumption:
    a. [1.0v] Report the mean and standard deviation of the accuracy and of the recall of the positive class for each classifier. Which is more stable regarding performance?
    b. [2.0v] Repeat the assessment of both models preprocessing the data with a Min-Max scaler (inside a pipeline). Explain the impact that this step has on the performance of each model.
2) Using a stratified 70-30 train-test split (`random_state=0`) and the Min-Max scaling of 1b) inside a pipeline, vary the number of neighbors of a kNN classifier using $k \in \{1, 5, 10, 15, 20, 25\}$. Additionally, for each k, train one classifier using uniform weights and distance weights.
    a. [1.0v] Plot the train and test accuracies for each model.

b. [1.5v] Explain the impact of increasing the number of neighbors on the generalization ability of the models, and the *trade-offs* between small and large values of k. Then suggest the optimal value of k. If you think the results do not justify choosing a single value, you may instead propose a range of values, explaining why a single value cannot be chosen.

3) [3.0v] Using the split of exercise 2, train the two pipelines of 1b) (Min-Max scaling followed by the kNN with $k = 5$, and Min-Max scaling followed by the Naïve Bayes), and explain some of their predictions with LIME (package `lime`, documentation and examples at https://github.com/marcotcr/lime).

LIME explains one prediction at a time. It generates many artificial observations around the observation to explain, asks the model to predict them, and fits a simple linear model that imitates the model around that point. The weights of this linear model are the explanation: a positive weight pushes the prediction towards the class being explained ("disease"), a negative weight away from it. By default, LIME splits each numeric variable into quartiles, so the explanation is given as conditions such as `oldpeak > 1.00`. The R^2 of the linear model, available as `exp.score`, measures how well it imitates the model (the *fidelity* of the explanation). LIME is random, so create a new explainer for each explanation, as in the function below; otherwise the results depend on the order in which the explanations are computed.

```
from lime.lime_tabular import LimeTabularExplainer

def explain(model, x, seed=0, num_samples=5000):
    explainer = LimeTabularExplainer(
        X_train, feature_names=features,
        class_names=["no disease", "disease"],
        mode="classification", random_state=seed)
    return explainer.explain_instance(
        x, model.predict_proba, num_features=5,
        num_samples=num_samples)

exp = explain(model, x)
exp.as_list()           # conditions and weights
exp.as_pyplot_figure()  # bar plot of the weights
exp.score               # fidelity of the explanation
```

`X_train` is the **unscaled** training data and `x` one row of the unscaled test data. Pass the `predict_proba` of the full pipelines, so that the explanations refer to the original values of the variables.

a. [1.0v] Select the first test patient with heart disease that is correctly classified by both models. Produce and plot the LIME explanation for each model. Do the two models rely on the same evidence? Naïve Bayes gives `fasting_blood_sugar` a much larger weight than the kNN: using the mean and variance that `GaussianNB` estimated for this variable in each class (attributes `theta_` and `var_` of the last step of the pipeline), explain why.

b. [1.0v] Select the first test patient that is a false negative (a patient with heart disease predicted as healthy) for at least one model. Using its LIME explanations, identify which conditions pushed the prediction towards the wrong class, and report the fidelity of each explanation. Can the explanation be trusted? How could a clinician looking at it have suspected the error?

c. [1.0v] Explanation stability: for the patient of 3a) and the kNN model, generate 10 LIME explanations with `seed` (the `random_state` of the explainer) from 0 to 9, using `num_samples=500` and then `num_samples=5000`. For each setting, report how often each variable appears in the top-5 and the standard deviation of its weight. Which parts of a single LIME explanation (the variables in the top-5, their order and their weights) can be trusted, and in which situations can they not?

4) [1.5v] Suppose you must deploy either kNN or Naïve Bayes in a hospital to support the diagnosis of heart disease. Choose one and give two reasons based on your results from exercises 1 to 3 (for example: recall, dependence on preprocessing, interpretability, computational cost). Then, using one result from exercise 3, discuss whether LIME explanations would be enough to justify deploying the model you did not choose.

Note: Use all optional parameters as default or justify any changes in your answers.